

Machine Learning I: Trees

HES 505 Fall 2025: Session 21

Matt Williamson

Updates

- Grading complete by end of week
- NO CLASS WEDS 5 Nov
- Final project descriptions!!

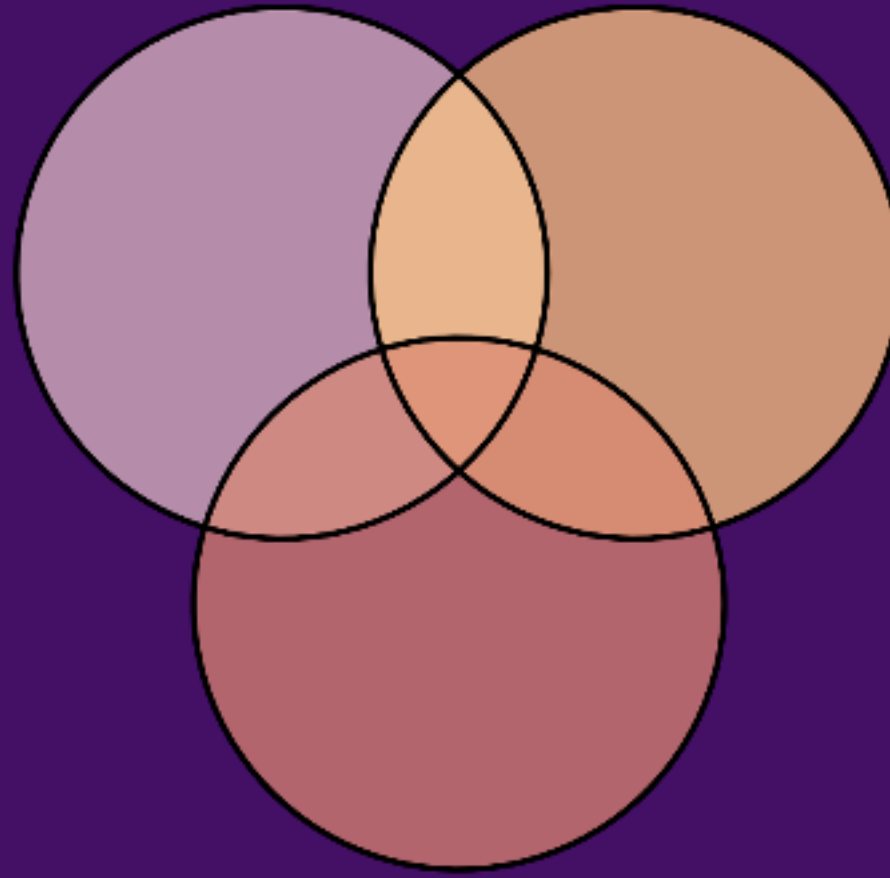
Statistical Learning

Objectives

- **Differentiate** supervised from unsupervised classification
- **Recognize** the linkage between statistical learning models and interpolation
- **Define** the key elements of tree-based classifiers
- **Articulate** the differences in statistical learning for spatial data

Description

Explanation



Prediction

Explaining Spatial Patterns

- **Cognitive Description:** collection ordering and classification of data
- **Cause and Effect:** design-based or model-based testing of the factors that give rise to geographic distributions
- **Systems Analysis:** describes the entire complex set of interactions that structure an activity

Explanation So Far

- **Deterministic Interpolation:** best explanation for variation is distance
- **Probabilistic Interpolation:** best explanation is covariance (plus spatial trends)

First Order Properties

$$z(\mathbf{x}) = \mu(\mathbf{x}) + \epsilon(\mathbf{x})$$

- Often we actually care about the “drivers” of $\mu(\mathbf{x})$
- Inference not prediction

Key Challenges

- Data volume
- Sampling designs
- Nonlinearity
- Autocorrelation

Statistical Learning

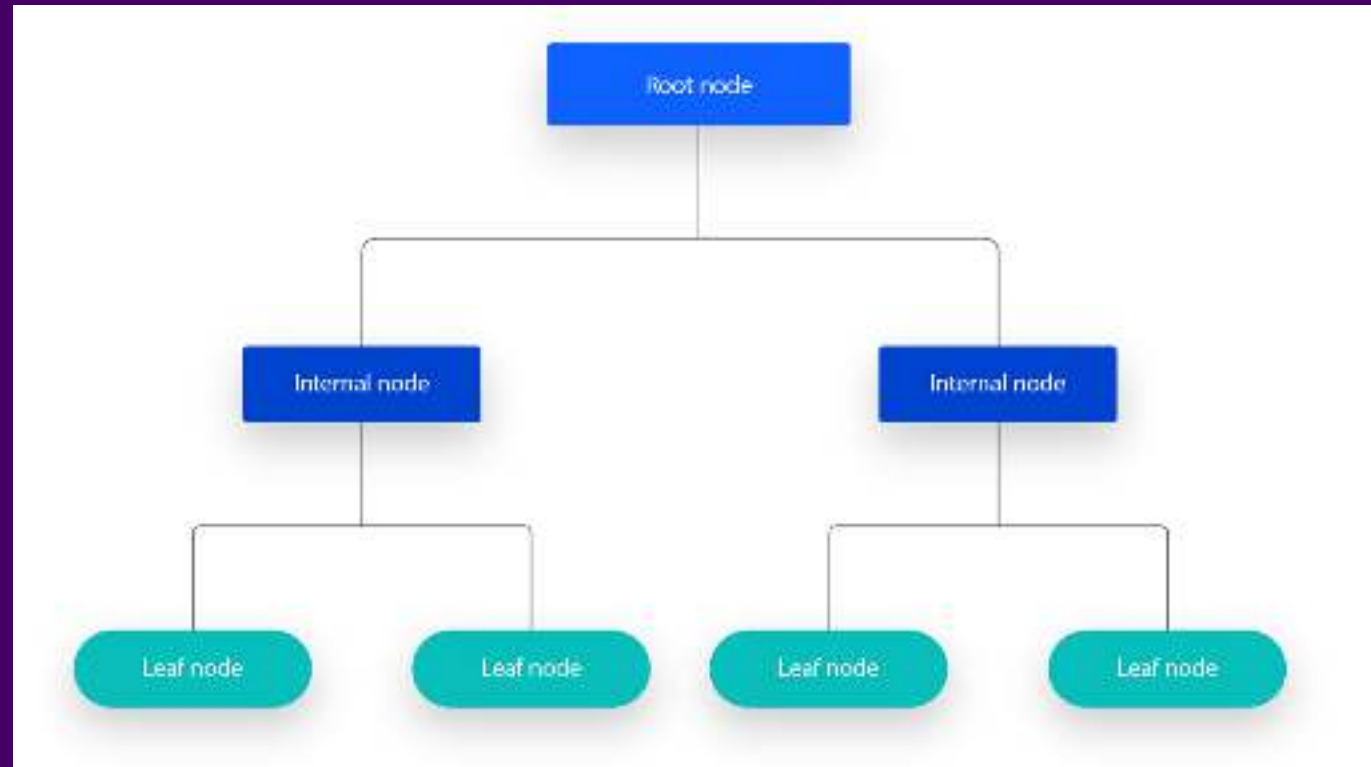
- “Machine Learning”
- Broad class of methods that rely on efficient, algorithmic learning
- Probabilities derived from “large” data (rather than theoretical distributions)
- Relax traditional statistical assumptions

Supervised Learning/Classification

- Goal is inference / prediction not grouping
- *Training* data: observations with known values used to generate model parameters
- *Validation* data: observations with known values used to tune hyperparameters prior to full model implementation
- *Testing* data: observations held out of initial model-fitting to determine accuracy

Classification Trees

- Use decision rules to segment the predictor space
- Series of consecutive decision rules form a 'tree'
- Terminal nodes (leaves) are the outcome; internal nodes (branches) the splits



Classification Trees

- Divide the predictor space (R) into J non-overlapping regions
- Every observation in R_j gets the same prediction
- *Recursive binary splitting*
- Maximize the information gained at each split
- Pruning and over-fitting

Benefits and drawbacks

Benefits

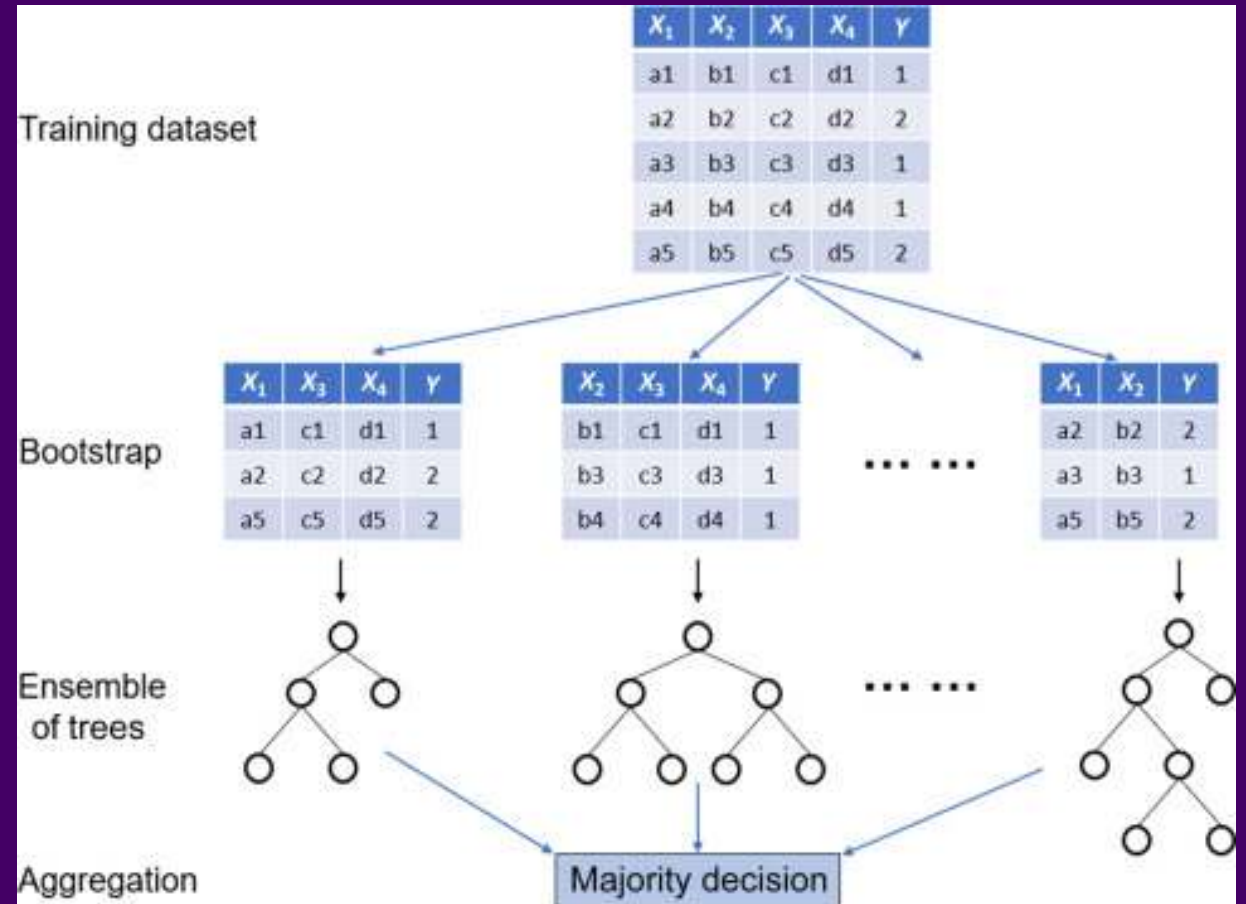
- Easy to explain
- Links to human decision-making
- Graphical displays
- Easy handling of qualitative predictors

Costs

- Lower predictive accuracy than other methods
- Not necessarily robust

Random Forests

- Grow 100(000s) of trees using bootstrapping
- Random sample of predictors considered at each split
- Avoids correlation amongst multiple predictions
- Average of trees improves overall outcome (usually)
- Lots of extensions



Sooooo Many Trees

- How to build ensembles?
- The number of and tuning of hyperparameters
- Boosting

Key Challenges with Tree-Based Methods

- No direct way to incorporate spatial autocorrelation
- Training Set qualities vs. Inferential conditions
- Cross-validation considerations

